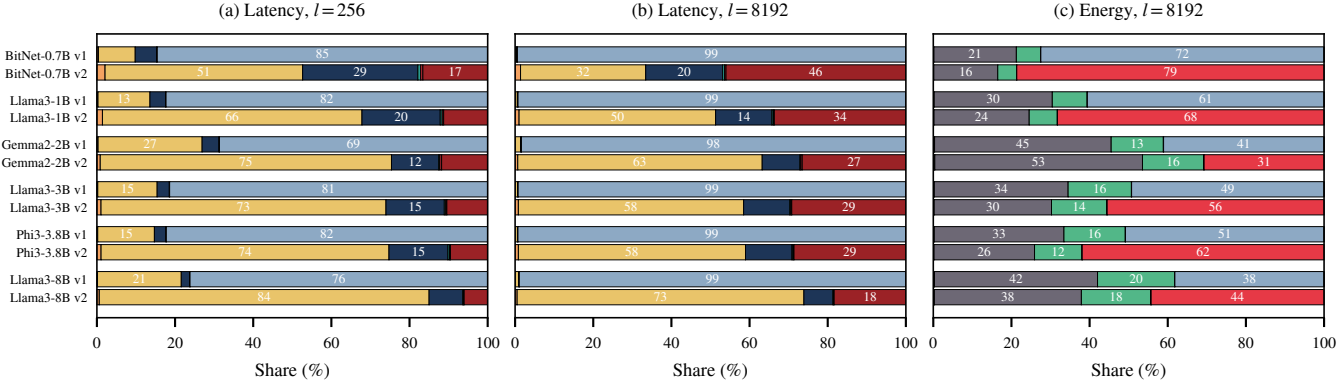
(a) Latency, $l = 256$
(b) Latency, $l = 8192$
(c) Energy, $l = 8192$
BitNet-0.7B v1
BitNet-0.7B v2
Llama3-1B v1
Llama3-1B v2
Gemma2-2B v1
Gemma2-2B v2
Llama3-3B v1
Llama3-3B v2
Phi3-3.8B v1
Phi3-3.8B v2
Llama3-8B v1
Llama3-8B v2
85
51
29
17
99
32
20
46
21
72
16
79
13
82
99
66
20
50
14
34
30
61
24
68
27
69
98
75
12
63
27
45
13
41
53
16
31
15
81
99
73
15
58
29
34
16
49
30
14
56
15
82
99
74
15
58
29
33
16
51
26
12
62
21
76
99
84
73
18
42
20
38
38
18
44
0
20
40
60
80
100
Share (%)
Latency (a, b):
Xbar+DAC+ADC
Communication
Buffer
Digital
Attn: systolic array (v1)
Attn: PIM xbar (v2)
Attn: PIM buffer (v2)
Energy (c):
Crossbar
DAC
ADC
Buffer
Other digital
Attn: systolic array (v1)
Attn: PIM (v2)